



Hadoop Cluster Setup Guide

Step-by-Step Cluster Deployment with ZooKeeper HA
Hadoop 3.4.2 | OpenJDK | Docker Environment

Document Type	Technical Setup Guide
Version	Hadoop 3.4.2
Cluster Nodes	5 Nodes (node01 – node05)
Environment	Docker Containers

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1. Environment Setup

1 Compose the Docker image

```
C:\Users\Mohamed\Desktop\DE\ITI\Projects\Hadoop\mini-hadoop-cluster\my-cluster>docker-compose up
time="2026-02-23T15:35:39+02:00" level=warning msg="C:\Users\Mohamed\Desktop\DE\ITI\Projects\Hadoop\mini-hadoop-cluster\my-cluster\docker-compose.yml: 'version' is obsolete"
[+] Running 5/0
  Container node05 Created 0.0s
  Container node01 Created 0.0s
  Container node02 Created 0.0s
  Container node03 Created 0.0s
  Container node04 Created 0.0s
Attaching to node01, node02, node03, node04, node05
```

2 Connect to the first node and download OpenJDK and Hadoop 3.4.2

```
C:\Users\Mohamed\Desktop\DE\ITI\Projects\Hadoop\mini-hadoop-cluster\my-cluster>docker exec -it node01 bash
root@node01:/# apt update && apt install -y openjdk-17-jdk wget nano net-tools
```

```
root@node01:/# wget -P /shared https://d1cdn.apache.org/hadoop/common/hadoop-3.4.2/hadoop-3.4.2.tar.gz
--2026-02-23 15:49:15-- https://d1cdn.apache.org/hadoop/common/hadoop-3.4.2/hadoop-3.4.2.tar.gz
Resolving d1cdn.apache.org (d1cdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to d1cdn.apache.org (d1cdn.apache.org)[151.101.2.132]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1065831750 (1016M) [application/x-gzip]
Saving to: '/shared/hadoop-3.4.2.tar.gz'
```

3 Extract the Hadoop archive with tar and move it to /opt/hadoop (on all nodes)

```
root@node01:/# tar -xzf /shared/hadoop-3.4.2.tar.gz -C /opt
root@node01:/# mv /opt/hadoop-3.4.2 /opt/hadoop
```

4 Set up the environment on all nodes

```
root@node02:/# echo 'export JAVA_HOME=/usr/lib/jvm/java-17-openjdk-amd64' >> ~/.bashrc
root@node02:/# echo 'export HADOOP_HOME=/opt/hadoop' >> ~/.bashrc
root@node02:/# echo 'export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin' >> ~/.bashrc
root@node02:/# source ~/.bashrc
```

 Environment variables were later improved by creating an `env_file` folder in the shared directory and configuring `docker-compose.yml` to use `env_file` — making changes easier and surviving `docker-compose` down restarts.

2. ZooKeeper Configuration

5 Attempt to download ZooKeeper (unstable version — do not use)

```
root@node01:/# wget -P /shared https://d1cdn.apache.org/zookeeper/zookeeper-3.8.4/apache-zookeeper-3.8.4-bin.tar.gz
--2026-02-24 10:58:07-- https://d1cdn.apache.org/zookeeper/zookeeper-3.8.4/apache-zookeeper-3.8.4-bin.tar.gz
Resolving d1cdn.apache.org (d1cdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to d1cdn.apache.org (d1cdn.apache.org)[151.101.2.132]:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-02-24 10:58:07 ERROR 404: Not Found.
```

 This version was unstable. Skip to Step 6 for the stable release.

6 Download the stable ZooKeeper version on one node

```
root@node01:/# wget -P /shared https://dlcdn.apache.org/zookeeper/zookeeper-3.8.6/apache-zookeeper-3.8.6-bin.tar.gz
--2026-02-24 10:59:45-- https://dlcdn.apache.org/zookeeper/zookeeper-3.8.6/apache-zookeeper-3.8.6-bin.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 16520101 (16M) [application/x-gzip]
Saving to: '/shared/apache-zookeeper-3.8.6-bin.tar.gz'

apache-zookeeper-3.8 100%[=====>] 15.75M 1.84MB/s in 9.0s

2026-02-24 10:59:55 (1.75 MB/s) - '/shared/apache-zookeeper-3.8.6-bin.tar.gz' saved [16520101/16520101]
```

7 Extract and move ZooKeeper to the correct folder (all nodes)

```
root@node01:/# tar -xzf /shared/apache-zookeeper-3.8.6-bin.tar.gz -C /opt
root@node01:/# mv /opt/apache-zookeeper-3.8.6-bin /opt/zookeeper
```

8 Set up the ZooKeeper environment (all nodes — same as Step 4)

```
root@node01:/# echo 'export ZOOKEEPER_HOME=/opt/zookeeper' >> ~/.bashrc
root@node01:/# echo 'export PATH=$PATH:$ZOOKEEPER_HOME/bin' >> ~/.bashrc
root@node01:/# source ~/.bashrc
```

9 Create data directory, assign unique ID to each ZooKeeper node, and set variables

```
root@node01:/# mkdir -p /opt/zookeeper/data
root@node01:/# echo "1" > /opt/zookeeper/data/myid
root@node01:/# cat > /opt/zookeeper/conf/zoo.cfg << EOF
> tickTime=2000
> initLimit=10
> syncLimit=5
> dataDir=/opt/zookeeper/data
> clientPort=2181
> server.1=node01:2888:3888
> server.2=node02:2888:3888
> server.3=node03:2888:3888
> EOF
```

```
root@node02:/# mkdir -p /opt/zookeeper/data
root@node02:/# echo "2" > /opt/zookeeper/data/myid
root@node02:/# cat > /opt/zookeeper/conf/zoo.cfg << EOF
> tickTime=2000
> initLimit=10
> syncLimit=5
> dataDir=/opt/zookeeper/data
> clientPort=2181
> server.1=node01:2888:3888
> server.2=node02:2888:3888
> server.3=node03:2888:3888
> EOF
```

```
root@node03:/# mkdir -p /opt/zookeeper/data
root@node03:/# echo "3" > /opt/zookeeper/data/myid
root@node03:/# cat > /opt/zookeeper/conf/zoo.cfg << EOF
> tickTime=2000
> initLimit=10
> syncLimit=5
> dataDir=/opt/zookeeper/data
> clientPort=2181
> server.1=node01:2888:3888
> server.2=node02:2888:3888
> server.3=node03:2888:3888
> EOF
```

10 Start ZooKeeper on all nodes

```
root@node01:/# zkServer.sh start
ZooKeeper JMX enabled by default
Using config: /opt/zookeeper/bin/../conf/zoo.cfg
Starting zookeeper ... STARTED
```

11 Check ZooKeeper status — verify 1 leader and 2 followers

```
root@node01:/# zkServer.sh status
ZooKeeper JMX enabled by default
Using config: /opt/zookeeper/bin/../conf/zoo.cfg
Client port found: 2181. Client address: localhost. Client SSL: false.
Mode: follower
```

```
root@node03:/# zkServer.sh status
ZooKeeper JMX enabled by default
Using config: /opt/zookeeper/bin/../conf/zoo.cfg
Client port found: 2181. Client address: localhost. Client SSL: false.
Mode: follower
```

```
root@node02:/# zkServer.sh status
ZooKeeper JMX enabled by default
Using config: /opt/zookeeper/bin/../conf/zoo.cfg
Client port found: 2181. Client address: localhost. Client SSL: false.
Mode: leader
```

 Seeing 1 leader and 2 followers confirms ZooKeeper quorum is healthy — we are on the right track.

3. Hadoop Configuration

12 Place Hadoop config files in the shared folder

Configuration files stored in /shared/config/: core-site.xml, hdfs-site.xml, mapred-site.xml, yarn-site.xml

13 Copy /shared/config/ to /opt/hadoop/etc/hadoop/ on all nodes

 /shared/config/ is accessible from all containers, making distribution straightforward.

```
C:\Users\Mohamed>docker exec -it node01 bash
root@node01:/# cp /shared/config/* /opt/hadoop/etc/hadoop/
```

14 Set JAVA_HOME path in the Hadoop environment file (all nodes)

```
root@node01:/# echo 'export JAVA_HOME=/usr/lib/jvm/java-17-openjdk-amd64' >> /opt/hadoop/etc/hadoop/hadoop-env.sh
```

15 Configure Hadoop daemons to run as root (required for Docker environments — all nodes)

```
root@node01:/# echo 'export HDFS_NAMENODE_USER=root' >> /opt/hadoop/etc/hadoop/hadoop-env.sh
root@node01:/# echo 'export HDFS_DATANODE_USER=root' >> /opt/hadoop/etc/hadoop/hadoop-env.sh
root@node01:/# echo 'export HDFS_JOURNALNODE_USER=root' >> /opt/hadoop/etc/hadoop/hadoop-env.sh
root@node01:/# echo 'export HDFS_ZKFC_USER=root' >> /opt/hadoop/etc/hadoop/hadoop-env.sh
root@node01:/# echo 'export YARN_RESOURCEMANAGER_USER=root' >> /opt/hadoop/etc/hadoop/hadoop-env.sh
root@node01:/# echo 'export YARN_NODEMANAGER_USER=root' >> /opt/hadoop/etc/hadoop/hadoop-env.sh
root@node01:/#
```

 This step is critical when running Hadoop inside Docker containers, since all processes run as root.

4. Cluster Initialization

16 Start the Journal Nodes (JN) on node01, node02, and node03

```
root@node02:/# hdfs --daemon start journalnode
WARNING: /opt/hadoop/logs does not exist. Creating.
```

17 Format the Active NameNode on node01, then bootstrap the Standby NameNode on node02

```
root@node01:/# hdfs namenode -format -clusterID mycluster
2026-02-24 15:29:19,776 INFO namenode.NameNode: STARTUP_MSG:
/*****
```

```
root@node02:/# hdfs namenode -bootstrapStandby
2026-02-24 15:30:26,246 INFO namenode.NameNode: STARTUP_MSG:
/*****
```

18 Format the ZooKeeper Failover Controller (ZKFC) on node01

```
root@node01:/# hdfs zkfc -formatZK
2026-02-24 15:29:39,600 INFO tools.DFSZKFailoverController: STARTUP_MSG:
/*****
```

19 Start zkServer.sh on node01, node02, and node03

```
root@node01:/# zkServer.sh start
ZooKeeper JMX enabled by default
Using config: /opt/zookeeper/bin/../conf/zoo.cfg
Starting zookeeper ... STARTED
```

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Start NameNodes (NN) and ZKFC on node01 & node02 — Start ResourceManager (RM) on node01

```
root@node01:/# hdfs --daemon start namenode
root@node01:/# hdfs --daemon start zkfc
root@node01:/# yarn --daemon start resourcemanager
root@node01:/# hdfs haadmin -getAllServiceState
```

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Start DataNodes (DN) and NodeManagers (NM) on node03, node04, and node05

```
root@node03:/# hdfs --daemon start datanode
root@node03:/# yarn --daemon start nodemanager
```

5. Verification & Testing

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Cluster is up and running — verify all services

```
root@node01:/# yarn rmdadmin -getAllServiceState
node01:8033          active
node02:8033          standby
root@node01:/# jps
1168 QuorumPeerMain
306 NameNode
1971 Jps
1655 ResourceManager
61 JournalNode
```

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Ingest data into HDFS

```
root@node04:/shared/data/jobs/mr_max_temperature# hdfs dfs -mkdir /test1
root@node04:/shared/data/jobs/mr_max_temperature# hdfs dfs -put /shared/data/data/weather_dataset/hourly/1901_1910.psv /test1
```

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Run a MapReduce job to validate the cluster end-to-end

```

root@node04:/shared/data/jobs/mr_max_temperature# hadoop jar mr_max_temperature.jar /test1/1901.1910.psv /output1
2026-02-25 00:28:20,063 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application
with ToolRunner to remedy this.
2026-02-25 00:28:20,112 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/root/.staging/job_1771952142034_0005
2026-02-25 00:28:20,541 INFO input.FileInputFormat: Total input files to process : 1
2026-02-25 00:28:20,735 INFO mapreduce.JobSubmitter: number of splits:2
2026-02-25 00:28:21,137 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1771952142034_0005
2026-02-25 00:28:21,143 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-02-25 00:28:21,518 INFO conf.Configuration: resource-types.xml not found
2026-02-25 00:28:21,519 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-02-25 00:28:21,768 INFO impl.YarnClientImpl: Submitted application application_1771952142034_0005
2026-02-25 00:28:21,812 INFO mapreduce.Job: The url to track the job: http://node01:8088/proxy/application_1771952142034_0005/
2026-02-25 00:28:21,815 INFO mapreduce.Job: Running job: job_1771952142034_0005
2026-02-25 00:28:33,028 INFO mapreduce.Job: Job job_1771952142034_0005 running in uber mode : false
2026-02-25 00:28:33,029 INFO mapreduce.Job: map 0% reduce 0%
2026-02-25 00:28:46,305 INFO mapreduce.Job: map 50% reduce 0%
2026-02-25 00:28:48,326 INFO mapreduce.Job: map 100% reduce 0%
2026-02-25 00:28:56,424 INFO mapreduce.Job: map 100% reduce 100%
2026-02-25 00:28:57,446 INFO mapreduce.Job: Job job_1771952142034_0005 completed successfully2026-02-25 00:28:57,605 INFO mapreduce.Job: Counters: 55

```

```

File System Counters
  FILE: Number of bytes read=21066441
  FILE: Number of bytes written=43073201
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=162809505
  HDFS: Number of bytes written=100
  HDFS: Number of read operations=11
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Killed map tasks=1
  Launched map tasks=2
  Launched reduce tasks=1
  Data-local map tasks=2
  Total time spent by all maps in occupied slots (ms)=24189
  Total time spent by all reduces in occupied slots (ms)=8007
  Total time spent by all map tasks (ms)=24189
  Total time spent by all reduce tasks (ms)=8007
  Total vcore-milliseconds taken by all map tasks=24189
  Total vcore-milliseconds taken by all reduce tasks=8007
  Total megabyte-milliseconds taken by all map tasks=24769536
  Total megabyte-milliseconds taken by all reduce tasks=8199168
Map-Reduce Framework

```

```

Map-Reduce Framework
  Map input records=1459246
  Map output records=1404429
  Map output bytes=18257577
  Map output materialized bytes=21066447
  Input split bytes=202
  Combine input records=0
  Combine output records=0
  Reduce input groups=10
  Reduce shuffle bytes=21066447
  Reduce input records=1404429
  Reduce output records=10
  Spilled Records=2808858
  Shuffled Maps=2
  Failed Shuffles=0
  Merged Map outputs=2
  GC time elapsed (ms)=299
  CPU time spent (ms)=26600
  Physical memory (bytes) snapshot=1005490176
  Virtual memory (bytes) snapshot=8298876928
  Total committed heap usage (bytes)=638582784
  Peak Map Physical memory (bytes)=406573056
  Peak Map Virtual memory (bytes)=276391168
  Peak Reduce Physical memory (bytes)=215740416
  Peak Reduce Virtual memory (bytes)=2774020096
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=162809303
File Output Format Counters
  Bytes Written=100

```

```

root@node04:/shared/data/jobs/mr_max_temperature# hdfs dfs -cat /output1/part-m-00001
cat: '/output1/part-m-00001': No such file or directory
root@node04:/shared/data/jobs/mr_max_temperature# hdfs dfs -ls /output1
Found 2 items
-rw-r--r-- 1 root supergroup          0 2026-02-25 00:28 /output1/_SUCCESS
-rw-r--r-- 1 root supergroup       100 2026-02-25 00:28 /output1/part-r-00000

```

```

root@node04:/shared/data/jobs/mr_max_temperature# hdfs dfs -cat /output1/part-m-00000
cat: '/output1/part-m-00000': No such file or directory
root@node04:/shared/data/jobs/mr_max_temperature# hdfs dfs -cat /output1/part-r-00000
1901 43.8
1902 45.6
1903 42.9
1904 42.4
1905 46.0
1906 42.8
1907 44.8
1908 45.0
1909 42.7
1910 46.1
root@node04:/shared/data/jobs/mr_max_temperature# _

```